

FITNOURISH AI

BSc (Hons) in Information Technology Specialization in Information Technology

Department of Information Technology

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FITNOURISH AI

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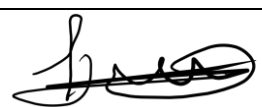
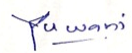
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ABSTRACT

Personalized dietary management remains a critical challenge in modern healthcare, particularly for individuals managing chronic conditions such as diabetes and hypertension. Existing systems typically address nutrition planning, meal generation, health risk assessment, and grocery purchasing in isolation, limiting their practical utility and real-world impact. This paper presents FitNourish.AI, a unified, AI-driven framework that integrates multi-disease adaptive nutrition modeling, dynamic meal planning with visual ingredient identification, predictive health analytics with longitudinal tracking, and travel-aware grocery optimization into a single closed-loop pipeline.

The proposed system comprises four interoperable modules. First, an XGBoost-based multi-output regression model [1] predicts personalized daily caloric and macronutrient requirements by incorporating physiological signals, wearable-derived activity metrics, and multi-disease indicators, with per-prediction transparency provided through SHAP-based Explainable AI [2, 3]. Second, a dynamic meal generator consumes nutritional targets to produce constraint-aware meal plans, supported by computer vision-based visual ingredient identification for real-time inventory management and shopping integration. Third, a predictive health alert and progress tracking module employs an XGBoost classifier [1] for three-tier risk stratification (Low, Medium, and High) based on key biomarkers including cholesterol, blood pressure, and glucose levels, complemented by a soft-voting ensemble model [4] for dietary pattern recommendation and a rule-based notification system with longitudinal biomarker dashboards. Fourth, a travel-aware grocery optimization module combines hybrid entity resolution strategies [5, 6] (incorporating Logistic Regression, Random Forest, and MLP-based product matching [7]) across local Sri Lankan retailers Keells and Cargills Food City, with a Travelling Purchaser Problem (TPP)-inspired heuristic [8, 9] that balances product price and travel cost to minimize total shopping expenditure.

Experimental results demonstrate strong performance across all modules. The nutrition prediction model achieved an R^2 of 0.957 with an MAE of 19.88, while the health risk classifier attained a test accuracy of 96.60% with a weighted F1-score of 0.97. The grocery

optimization module achieved a cost reduction of 12–25% across evaluated shopping baskets. All modules share a snapshot-based, reproducible data pipeline [10, 11] and communicate through standardized APIs, enabling seamless end-to-end user journeys from health profiling and nutrition prediction through meal generation, grocery optimization, and progress tracking. Collectively, these results confirm that FitNourish.AI effectively bridges the gap between computational health intelligence and real-world decision support, offering a scalable, interpretable, and ethically grounded solution for personalized dietary and health management.

Keywords: Personalized Nutrition, XGBoost, Explainable AI, Health Risk Prediction, Meal Planning, Visual Ingredient Identification, Grocery Optimization, Entity Resolution, Travelling Purchaser Problem, Wearable Integration, Multi-Disease Modeling, Ensemble Learning

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
BMI	Body Mass Index
CNN	Convolutional Neural Network
GPU	Graphics Processing Unit
ML	Machine Learning
SLIIT	Sri Lanka Institute of Information Technology
YOLO	You Only Look Once
LSTM	Long Short Term Memory
MAP	Mean Average Precision
MAPE	Mean Absolute Percentage Error

1 INTRODUCTION

1.1 Background

The rapid proliferation of chronic non-communicable diseases (NCDs) such as type 2 diabetes, hypertension, and cardiovascular disorders has emerged as one of the most pressing public health challenges of the twenty-first century. Dietary behavior and nutritional intake are widely recognized as primary modifiable risk factors governing the onset and progression of these conditions [12]. Despite growing awareness, individuals managing multiple concurrent chronic conditions face significant difficulties in translating generalized clinical dietary guidelines into actionable, day-to-day nutritional decisions that account for their unique physiological profiles, activity levels, and health trajectories. This gap between clinical recommendation and practical implementation underscores the urgent need for intelligent, personalized health management systems capable of operating in real-world consumer environments.

Advances in machine learning, ensemble modeling, and explainable artificial intelligence have opened new avenues for delivering precision nutrition at scale [13, 4]. Supervised learning frameworks particularly gradient boosting methods such as XGBoost [1] have demonstrated exceptional predictive performance on structured tabular health data, enabling the modeling of complex, non-linear relationships between physiological signals and nutritional requirements. Simultaneously, the widespread adoption of wearable health devices and mobile health platforms has made continuous, real-time acquisition of biometric and activity data increasingly accessible to everyday users, creating unprecedented opportunities for data-driven, personalized dietary management [12].

However, despite these technological advances, existing digital health solutions remain largely fragmented. Nutrition tracking applications operate independently of health risk assessment tools; meal planning platforms rarely incorporate real-time physiological feedback; and grocery purchasing decisions are seldom integrated with either nutritional goals or health status. This siloed architecture limits the practical utility of individual applications and places an undue cognitive burden on users who must manually coordinate information across multiple disconnected platforms. Furthermore, the majority of existing

systems are developed and validated within the context of well-resourced healthcare ecosystems, with limited attention to developing-world retail environments and local consumer contexts [10].

The integration of recommendation systems [14, 15, 16], optimization theory [17, 18], and geospatial computation [19, 20] into health-oriented platforms represents a promising but underexplored direction. In particular, grocery purchasing a daily behavior with direct consequences for dietary quality has received relatively little attention in AI-driven health optimization research. Product matching across heterogeneous retail datasets presents non-trivial entity resolution challenges [5, 21, 6], while multi-store purchasing decisions introduce combinatorial complexity analogous to the Travelling Purchaser Problem (TPP) [8, 9], a well-studied variant of routing and purchasing optimization that seeks to minimize the combined cost of goods and travel across multiple vendor locations.

To address these limitations, this paper presents FitNourish.AI, a unified, closed-loop AI framework that integrates four interoperable intelligent modules into a single cohesive health optimization pipeline. The system encompasses: (1) an AI-powered multi-disease adaptive nutrition prediction module; (2) a dynamic meal generator supported by visual ingredient identification; (3) a predictive health alert and longitudinal progress tracking module; and (4) a travel-aware smart grocery list optimization and nearest store recommendation module. Together, these components enable a seamless end-to-end user journey from initial health profiling and personalized macronutrient estimation, through constraint-aware meal planning and visual inventory management, to risk-stratified health monitoring and cost-optimized grocery procurement.

The nutrition prediction component employs a MultiOutputRegressor wrapper around XGBoost [1] to simultaneously predict daily caloric, protein, carbohydrate, and fat requirements from an enriched feature space combining demographic, physiological, and wearable-derived activity signals. Model transparency is ensured through SHAP-based Explainable AI [2, 3], providing both global feature importance rankings and local per-prediction explanations a critical requirement for health-sensitive recommendations where interpretability directly influences user trust and clinical credibility. The health risk stratification module applies a rigorously validated XGBoost classifier to categorize users

into Low, Medium, and High risk tiers based on key clinical biomarkers, while a soft-voting ensemble model [4, 22, 23] generates personalized dietary pattern recommendations. A rule-based notification engine and longitudinal biomarker dashboard further support sustained behavioral engagement and health progress monitoring.

The grocery optimization module addresses the practical challenge of translating nutritional recommendations into cost-effective purchasing decisions within a real-world retail context. Product data is collected from two major Sri Lankan supermarket chains Keells and Cargills Food City through automated web scraping and API integration, and stored in a reproducible snapshot-based architecture [10, 11]. A hybrid entity resolution pipeline combining canonical mapping, rule-based heuristics, and ML classifiers including Logistic Regression, Random Forest [7], and MLP resolves product identities across heterogeneous retailer datasets [5, 24, 6]. Travel cost modeling leverages the Haversine formula [19] and Google Distance Matrix API [25] to enable a TPP-inspired multi-store optimization heuristic [8, 9] that minimizes total shopping expenditure by intelligently balancing product prices against travel costs.

The primary contributions of this research are as follows:

- A novel integrated AI framework that unifies nutrition prediction, meal generation, health risk monitoring, and grocery optimization into a single closed-loop personalized health management system.
- An XGBoost-based multi-output regression model enriched with wearable-derived physiological and activity features and explained through SHAP values, achieving an R^2 of 0.957 and MAE of 19.88 for macronutrient prediction.
- A three-tier health risk classification system achieving 96.60% test accuracy and a weighted F1-score of 0.97, complemented by an ensemble dietary recommendation model and longitudinal tracking dashboard.
- A travel-aware grocery optimization module employing hybrid entity resolution and a TPP-inspired heuristic, yielding cost reductions of 12–25% across evaluated shopping baskets in a developing-world retail context.

- A modular, reproducible system architecture based on standardized APIs, snapshot-based data management, and serialized model artifacts, supporting scalability, maintainability, and real-world deployability.

1.2 Literature Survey

Personalized nutrition and mobile health (m-Health) applications have advanced rapidly with the integration of wearable sensors, smartphones, and machine learning. Early systems relied on rule-based or static equations for calorie and macronutrient targets. Recent work has shifted toward data-driven, multi-disease-aware models. Liu et al. [26] developed an m-Health framework that leverages smartphone data and machine learning to deliver context-aware nutrition services. Complementary studies have explored hybrid deep-learning architectures for glycemic control [27] and reinforcement-learning approaches for dynamic meal planning [28]. Omics-integrated frameworks further personalize recommendations while addressing ethical and privacy concerns [29]. Despite these advances, most solutions remain isolated. They generate static targets without incorporating real-time activity data from wearables or forward-looking health risk forecasts [26], [30], [29].

Health forecasting has emerged as a critical extension of m-Health monitoring. Machine learning classification models have been applied to stratify health risks using physiological biomarkers and lifestyle attributes [31], [32]. However, these forecasting modules are rarely coupled with actionable downstream decisions such as personalized grocery procurement. This leaves users with risk alerts but no practical mitigation pathways [33]–[32].

Travel-aware grocery recommendation and sustainable last-mile optimization represent a parallel research stream. Context-aware systems such as SmartGrocer [34] and multi-store cost-optimization apps integrate price comparison with user preferences and geolocation. More advanced logistics models formulate the problem as variants of the Travelling Purchaser Problem (TPP) or multi-echelon vehicle routing, incorporating perishability and travel costs [33], [35]–[37]. Entity-resolution techniques address catalogue heterogeneity across retailers [5], [6]. These optimization frameworks, however, operate independently of individual nutritional needs or predicted health risks. As a result, they produce plans that may satisfy cost or convenience but fail to align with disease-specific dietary targets [7], [38].

A growing body of research focuses on AI-powered dynamic meal generation and visual ingredient identification to bridge the gap between nutritional recommendations and practical meal planning. Several studies have proposed intelligent food recommendation systems that generate personalized meal plans based on user health profiles and pathological test results [39], [31]. Advanced deep learning approaches for food recognition and freshness assessment have also gained attention. Ghosh et al. [40] proposed a hybrid CNN-LSTM model that achieved 98.9% accuracy in classifying fresh versus rotten fruits by combining spatial features with pseudo-temporal ripening dynamics. Other works have explored transfer learning and ensemble methods [38], ingredient-guided attention networks [41], and multispectral imaging with CNNs [42] to improve food image recognition accuracy and caloric estimation. Studies evaluating commercial nutrition apps further highlight the advantages of AI-enabled food image recognition over manual logging, while revealing persistent challenges in accuracy for culturally diverse and mixed dishes [43], [44].

Table I provides a comparative overview of representative systems across the four core dimensions of this study.

Literature	Multi-Disease Nutrition	Health Forecasting	Travel-Aware Grocery Optimization	AI-Powered Meal Generation & Visual Identification	Unified Closed-Loop
Liu et al. [26]	Partial	No	No	Limited	No
Hybrid CNN-RNN & Omics [29], [27]	Yes	Limited	No	Partial	No
Forecasting models [33], [32]	No	Yes	No	No	No
SmartGrocer & TPP models [34], [35]–[37]	No	No	Yes	No	No

Diet-Right & ML recommenders [39]	Yes	No	No	Yes	No
Food recognition & freshness models [38], [40]–[42]	No	No	No	Yes	No
This work (FitNourish.AI)	Yes (XGBoost)	Yes (XGBoost)	Yes (MLP + TPP)	Yes (Dynamic Meal + Visual Identifier)	Yes

Table 1-1 Comparative Analysis of State-of-the-Art Systems

1.3 Research Gap

While prior studies have made significant progress in isolated domains such as personalized nutrition [26], [29], [28], [39], [27], health forecasting [31], [33], [32], travel-aware grocery logistics [34], [35]–[37], and AI-powered meal generation with visual food recognition [43], [38], [40]–[44], no existing solution integrates these four layers into a single closed-loop pipeline. Static nutrition calculators ignore real-time wearable-derived activity data [26], [29]. Forecasting models rarely translate risk signals into procurement actions [33], [32]. Grocery optimizers lack disease-aware nutritional constraints [7], [38]. Similarly, dynamic meal generators and food recognition systems seldom connect recommendations with real-world grocery availability and travel considerations. This fragmentation is particularly pronounced in emerging markets such as Sri Lanka, where local dietary patterns, retailer catalogues (Keells and Cargills), and urban mobility constraints must be jointly addressed.

1.4 Research Problem

Despite the rapid development of artificial intelligence, machine learning, wearable technologies, and mobile health applications, existing nutrition and grocery-related systems remain fragmented. Most available systems focus on isolated functions such as calorie tracking, meal recommendation, health risk prediction, food recognition, or grocery price comparison. Therefore, users often need to depend on multiple disconnected applications to manage diet, monitor health, plan meals, and purchase groceries. This creates a gap between personalized health recommendations and practical daily food-related decisions [12], [14], [15], [26], [38], [43], [44].

This problem is more significant for individuals with chronic non-communicable diseases such as diabetes, hypertension, and cardiovascular conditions because they require dietary decisions that adapt to health status, activity levels, physiological indicators, and long-term progress. However, many existing nutrition systems provide generalized or static recommendations and do not sufficiently consider wearable-derived activity data, multi-disease conditions, explainable prediction, or future health risks [26], [27], [28], [29], [31], [43], [44], [52].

Another major limitation is the lack of integration between personalized nutrition recommendations and real-world grocery purchasing. Even after receiving meal plans or macronutrient targets, users must manually identify ingredients, compare supermarket prices, check availability, and select suitable stores. Current grocery and recommendation systems mainly consider price, preference, or convenience, while often ignoring disease-related dietary constraints, personalized nutrition needs, health risk levels, and local food availability [14], [15], [16], [33]- [37].

In addition, grocery optimization in real retail environments is technically challenging because supermarket data often contains inconsistent product names, unit sizes, categories, prices, and availability. Accurate product matching and price comparison therefore require data integration, data cleaning, entity resolution, and similarity-based matching. Practical store recommendations must also consider travel distance, travel cost, and possible multi-store purchasing, rather than selecting only the cheapest product [5], [6], [10], [11], [21], [24], [32], [37], [50].

Therefore, the research problem addressed in this study is the absence of a unified AI-powered platform that integrates multi-disease adaptive nutrition prediction, dynamic meal planning, visual ingredient identification, predictive health monitoring, and travel-aware grocery optimization. FitNourish.AI aims to address this gap by connecting personalized health profiling, nutrition prediction, meal generation, grocery price comparison, and nearest store recommendation within one interoperable system suitable for the Sri Lankan retail and health management context [1], [2], [3], [17], [18], [19], [20], [25], [47], [52].

1.5 Objectives

1.5.1 Main objectives

The main objective of this research is to design, develop, and evaluate an AI-powered Smart Grocery and Personalized Health Optimization System that supports users in making personalized, health-aware, and economically efficient food-related decisions. The proposed system aims to integrate personalized nutrition prediction, dynamic meal planning, visual ingredient identification, smart grocery list optimization, nearest store recommendation, predictive health alerts, and progress tracking into a unified intelligent platform.

Unlike conventional nutrition applications or grocery price comparison systems that operate as isolated solutions, this research seeks to bridge the gap between personalized health optimization and practical grocery purchasing support. The system is designed to analyse user health profiles, demographic details, physiological data, activity patterns, dietary goals, supermarket product information, product availability, store locations, and price variations to generate recommendations that are both nutritionally suitable and practically affordable. By combining machine learning, data preprocessing, product matching, optimization techniques, and health risk prediction, the proposed system aims to provide a holistic approach to daily nutrition and grocery decision-making.

Furthermore, the main objective emphasizes the development of a modular, scalable, and interoperable architecture in which each AI-powered component contributes to the overall system. The nutrition prediction module estimates personalized calorie and macronutrient requirements, while the meal generation and ingredient identification module supports meal planning and food recognition. The grocery optimization module compares products across supermarkets and recommends cost-effective purchasing options by considering product prices and travel-related factors. The health alert and progress tracking module monitors health patterns, predicts risk levels, recommends dietary plans, and provides reminders. Together, these components aim to deliver an end-to-end intelligent system that improves health awareness, supports better dietary decisions, and enhances cost-effective grocery shopping.

1.5.2 Specific objectives

- 1 To design and implement an AI-powered nutrition prediction module that estimates personalized daily calorie and macronutrient requirements based on user demographic, physiological, health-related, and activity-based data.
- 2 To develop a dynamic meal generation module that provides personalized meal suggestions according to user nutritional requirements, dietary goals, health conditions, and system-generated nutrition outputs.
- 3 To integrate a visual ingredient identification component that supports ingredient recognition and assists users in connecting available food items with meal planning and grocery recommendations.
- 4 To develop a smart grocery list optimization module that matches products across multiple supermarket sources using data preprocessing, product matching, and machine learning-based classification techniques.
- 5 To implement cross-store price comparison and availability analysis to help users identify suitable grocery products based on current supermarket data.
- 6 To develop a nearest and cost-effective store recommendation mechanism that considers product prices, item availability, store location, travel distance, and travel-related cost.

- 7 To implement a predictive health alert and progress tracking module that analyses user health records, predicts risk levels, provides dietary recommendations, and monitors long-term health progress.
- 8 To integrate all AI-powered modules into a unified, modular, scalable, and interoperable system that supports personalized health optimization and practical grocery decision-making.
- 9 To evaluate the proposed system using appropriate testing methods, model evaluation metrics, and system-level validation to ensure accuracy, usability, efficiency, and reliability.

2 METHODOLOGY

The methodology of the system integrates four independently developed AI modules into a cohesive, end-to-end intelligent platform. The system enables users to receive personalized nutrition targets, generate dynamic meal plans from food images, optimize grocery lists with multi-store price and travel cost awareness, and track health progress with predictive alerts.

The four modules are:

- AI-powered Multi-Disease Adaptive Nutrition and Personalized Health Optimization
- AI-powered Dynamic Meal Generator and Visual Ingredient Identifier
- AI-powered Smart Grocery List Optimization and Nearest Store Recommendation
- AI-powered Predictive Health Alert Progress Tracking System

The overall architecture follows a modular, layered, data-driven pipeline with shared design principles. These include immutable snapshot-based data storage for reproducibility, rigorous preprocessing, hybrid machine learning and optimization techniques, standardized data interfaces, and strong emphasis on real-world practicality, scalability, and ethical considerations.

2.1 AI-powered Multi-Disease Adaptive Nutrition and Personalized Health Optimization

2.1.1 Key Research Pillars

This module is built upon five core pillars:

- Machine Learning in Healthcare – for learning complex patterns from multi-variable user data.

- Regression Models – suitable for continuous numerical outputs (calories, protein, carbs, fat).
- Ensemble Learning via XGBoost – chosen for strong performance on tabular data, regularization, and feature interaction modeling.
- Multi-Output Regression – enables simultaneous prediction of four interrelated nutritional targets.
- Explainable AI (SHAP) – delivers global and local interpretability for health recommendations.

2.1.2 Data Collection and Dataset Description

The dataset captures demographic information (age, gender, height, weight, goal), physiological and health indicators (resting and average heart rate, diabetes, hypertension, stress score), and activity-related features (steps, active minutes, calories burned). The targets are daily calorie requirement, protein, carbohydrate, and fat intake per day. Data is stored in structured CSV format with clear input-output separation.

2.1.3 Data Preprocessing and Feature Engineering

Preprocessing includes median imputation for numerical features, logical defaults for binary variables, one-hot encoding for categoricals, Z-score standardization, and removal of records missing targets. Feature engineering creates derived variables such as BMI, activity ratio, heart efficiency, steps per minute, and stress impact.

2.1.4 Model Development, Training, Prediction, and Evaluation

A MultiOutputRegressor wrapping XGBoost is trained on the prepared data and persisted using Joblib. The prediction workflow applies identical preprocessing and feature engineering steps before inference and optionally computes SHAP explanations.

Evaluation uses MAE, RMSE, R^2 , and cross-validation, with per-output metrics reported separately. This ensures high accuracy, reliability, and generalization.

2.2 AI-powered Dynamic Meal Generator and Visual Ingredient Identifier

2.2.1 Key Research Pillars

The module combines computer vision and mathematical optimization:

- Deep Learning and CNNs for hierarchical image feature extraction.
- Transfer Learning with EfficientNetV2B0 (ImageNet pre-trained, fine-tuned on food data).
- Multi-label Classification (Sigmoid + Binary Crossentropy) for 75 ingredients.
- Residual Learning (skip connections) for deep network training.
- Integer Linear Programming (ILP) via PuLP for meal optimization.
- Portion-Independent Feature Engineering and the 4-4-9 Caloric Rule (4 kcal/g protein and carbs, 9 kcal/g fat) for volume-invariant prediction and validation.

2.2.2 Data Source and Preprocessing

The Nutrition5k dataset (~5,000 realistic meals) provides multi-angle images, exact per-ingredient mass, total dish mass, and USDA nutritional ground truth.

Preprocessing includes:

- Selection of top 75 ingredients by mass (>5% threshold).
- Image standardization to 320×320 pixels (LANCZOS resampling).
- Parallel processing with ThreadPool.
- Data augmentation (rotations $\leq 45^\circ$, flips, brightness adjustment).
- Conversion of nutrients to portion-independent density ratios.
- train/test split.

2.2.3 System Development Workflow

- Visual Analysis Engine: EfficientNetV2B0 with transfer learning and multi-label Sigmoid output.
- Nutrient Estimation Engine: Portion-independent regression with persisted MinMaxScaler.
- Personalization and Optimization Engine: ILP minimizes absolute caloric deviation using binary dish variables and auxiliary over/under-target variables, subject to hard constraints (mandatory ingredients, meal counts). Achieved 84.58% planning success rate.

The 4-4-9 rule provides secondary validation and caloric distribution insights.

2.2.4 Requirements, Tools, and Technologies

- Functional requirements include image upload, multi-ingredient identification, nutritional reporting, health-profile suggestions, user profiles, and history.
- Non-functional requirements include $\geq 90\%$ accuracy, $\leq 10s$ response time, cross-platform support, scalability, and security.
- Tools used are Python, React Native, FastAPI, MongoDB, TensorFlow/PyTorch, OpenCV, PuLP, Figma, VS Code, and GitHub.

2.2.5 Testing and Implementation

Models were evaluated on unseen test dishes using tolerance-based accuracy and MSE. The full pipeline (image $\rightarrow$ ingredients $\rightarrow$ nutrients $\rightarrow$ optimized meal plan) was validated end-to-end with results serialized to JSON.

2.3 AI-powered Smart Grocery List Optimization and Nearest Store Recommendation

2.3.1 Key Research Pillars

- Entity Resolution and Product Matching (hybrid rule-based + ML + canonical mapping).
- Machine Learning Models (Logistic Regression, Random Forest, MLP).
- Optimization Theory and Travelling Purchaser Problem (TPP).
- Distance Calculation (Haversine + Google Distance Matrix API).

2.3.2 Data Collection, Storage, and Preprocessing

Data scraped from Keells (API) and Cargills (Playwright). Raw snapshots stored in JSONL/CSV and PostgreSQL with time-stamped tables (snapshot-based architecture). Preprocessing includes text normalization, tokenization, unit standardization, noise removal, canonical grouping, rule-based categorization, and comprehensive feature engineering (string similarities, token overlap, size ratios, etc.).

2.3.3 Product Matching, Price Comparison, and Optimization

Hybrid matching strategy (canonical → rules → Random Forest). Price and availability aggregation across stores. Travel cost modeling using Haversine for speed and Google API for accuracy. Total cost minimization (items + travel). Single-store and TPP-inspired heuristic multi-store optimization (top-K filtering, greedy allocation, net-savings decision rule).

2.3.4 Implementation, Testing, and Tools

The dataset was split 60/20/20 for the matching model. Tools used include Flask, PostgreSQL with Alembic, scikit-learn, RapidFuzz, NumPy/Pandas, and Google Distance Matrix API. Evaluation focuses on precision, recall, F1-score, and system efficiency.

2.4 AI-powered Predictive Health Alert Progress Tracking System

2.4.1 System Architecture

Layered RESTful client-server architecture with four modules: Data Collection, XGBoost Risk Prediction (Low/Medium/High), Voting Ensemble Dietary Recommendation (Low-Carb, Low-Sodium, Balanced, Mediterranean), and Notification and Tracking Dashboard.

2.4.2 Datasets and Preprocessing

- Risk dataset: 5,000 records with 19 engineered features (binary/ordinal/one-hot encoding, scaler on train only).
- Diet dataset: 1,000 records, SMOTE-balanced, with 25 engineered features (risk scores, interactions, polynomials).

2.4.3 Model Development and Evaluation

XGBoost was selected after 5-fold CV (96.07% accuracy). The Voting Ensemble achieved 66.94% test accuracy. Strict prevention of data leakage was maintained.

2.4.4 Notification, Tracking, Requirements, Ethics, and Security

Rule-based pre-meal reminders and an interactive biomarker dashboard with risk overlays are used. Comprehensive functional and non-functional requirements were defined.

Ethical measures include informed consent, clinical disclaimers, and fairness audits. Security uses JWT, TLS, password hashing, and data minimization.

2.5 Overall System Integration and Reproducibility

All modules share immutable snapshot storage, consistent preprocessing pipelines, and serialized model artifacts. Standardized nutritional vectors and user profiles enable seamless data flow from health profiling to personalized nutrition targets, image-based meal generation, grocery optimization, and progress tracking with alerts. This integration creates a complete, practical, and reproducible intelligent health ecosystem.

2.6 Consolidated Tools and Technologies

Common Stack: Python, virtual environments, GitHub.

Module-Specific:

- Nutrition: XGBoost, scikit-learn, SHAP, Joblib, Pandas, NumPy.
- Meal Generator: EfficientNetV2B0 (TensorFlow/PyTorch), PuLP, OpenCV, React Native, FastAPI, MongoDB.
- Grocery: Flask, PostgreSQL + Alembic, Playwright, Requests, RapidFuzz, Google Distance Matrix API.
- Health Tracking: XGBoost, Voting Ensemble, StandardScaler, Android Notifications.

2.7 Project Requirements

This section presents the consolidated project requirements for the entire FitNourish AI: AI-Powered Smart Grocery and Personalized Health Optimization System. These requirements integrate the functional needs of all four modules (Personalized Nutrition Prediction, Dynamic Meal Generator with Visual Ingredient Identifier, Smart Grocery

Optimization, and Predictive Health Alert & Progress Tracking) into a unified specification.

2.7.1 Functional requirements

- 1 Collection and storage of user profile data, including demographic, physiological, health-related, lifestyle, and activity-based information, has been implemented to support personalized nutrition and health analysis.
- 2 Prediction of personalized daily calorie and macronutrient requirements, including protein, carbohydrate, and fat intake, using machine learning-based nutrition prediction has been achieved.
- 3 Generation of personalized meal suggestions based on user nutritional requirements, dietary goals, health conditions, and system-generated nutrition outputs has been implemented.
- 4 Visual ingredient identification to recognize food ingredients and support meal planning, inventory awareness, and grocery recommendation has been included as part of the integrated system.
- 5 Collection and storage of grocery product data from multiple supermarket sources, including product name, category, unit size, price, availability, and store details, has been achieved.
- 6 Preprocessing and transformation of raw grocery data, including product name normalization, unit standardization, duplicate removal, categorization, and feature preparation, has been implemented.
- 7 Accurate product matching across different supermarkets using a hybrid approach that combines machine learning, rule-based similarity methods, and canonical mapping has been achieved.

- 8 Price comparison and availability analysis of grocery products across multiple supermarkets has been implemented to support cost-aware purchasing decisions.
- 9 Generation of optimized grocery purchasing recommendations, including cheapest store, nearest store, and multi-store purchasing plans based on item cost, availability, distance, and travel cost, has been successfully implemented.
- 10 Prediction of user health risk levels using health records, clinical indicators, lifestyle factors, and machine learning-based classification has been implemented.
- 11 Generation of personalized dietary recommendations based on user health risk level, clinical information, and lifestyle data has been achieved.
- 12 Reminder notification functionality for meal timing, pre-meal alerts, and health-related actions has been implemented to support user adherence and engagement.
- 13 Health progress tracking through user-entered records such as weight, blood pressure, cholesterol, glucose level, exercise, sleep, and diet information has been implemented.
- 14 Dashboard-based visualization of health progress, biomarker trends, risk status, and historical records has been included to support continuous monitoring.
- 15 Integration of nutrition prediction, meal generation, grocery optimization, store recommendation, and health tracking modules through standardized system flows and APIs has been achieved.

2.7.2 Non-functional requirements

- 1 **Accuracy:** The system shall ensure reliable predictions and recommendations, achieving $\geq 75\%$ food/ingredient identification accuracy, $\geq 95\%$ health risk prediction accuracy, and strong R^2 , precision, recall, and F1-score results.

- 2 **Performance:** Nutritional report generation and image analysis shall be completed within 10 seconds, health risk prediction and dietary recommendation within 300 milliseconds, and grocery optimization within 2 seconds.
- 3 **Usability:** The system shall provide a simple, clear, and user-friendly mobile interface for users with no technical or clinical background.
- 4 **Scalability:** The system shall support future expansion for more users, supermarket data, grocery products, health records, wearable data, and recommendation requests.
- 5 **Reliability and Availability:** The system shall maintain at least 99% backend uptime and 98% notification delivery success under normal conditions.
- 6 **Security and Privacy:** The system should protect user data using JWT authentication, TLS encryption, password hashing, input validation, rate limiting, role-based access control, and data minimization.
- 7 **Interoperability:** The system shall communicate smoothly between the mobile app, backend, database, machine learning models, grocery optimization module, and health tracking module through standardized APIs.
- 8 **Compatibility:** The system shall support Android API level 26 or higher and iOS devices, with possible wearable health data integration.
- 9 **Reproducibility:** The system shall maintain version-controlled models, preprocessing pipelines, serialized artifacts, structured datasets, and snapshot-based supermarket data.

2.7.3 User requirements

End users (primarily health-conscious individuals, fitness enthusiasts, and patients managing chronic conditions) shall be able to:

1. Easily register and create/update a personal health profile with minimal effort.

2. Capture meal images using the phone camera or upload from gallery for instant nutritional analysis.
3. Receive clear, understandable nutritional breakdowns, health risk alerts, and actionable dietary recommendations.
4. Get optimized, personalized daily meal plans that consider their health goals and preferences.
5. Generate smart grocery lists with nearest/best-value store recommendations, including multi-store options when beneficial.
6. View historical data and progress trends through an interactive dashboard.
7. Receive timely, relevant push notifications for meals and health alerts.
8. Have full visibility and control over their personal data, with the ability to export or delete records.
9. Use the application confidently knowing that recommendations are supportive tools and not substitutes for professional medical advice (clear disclaimers provided).

2.7.4 System Requirements

2.7.4.1 Hardware requirements

1. Smartphones: A mobile device with a camera resolution of at least 12 MP is required.
2. Server: A dedicated server with GPU support is necessary for training and running the deep learning models.
3. Internet: Stable and fast internet connectivity is required for data upload and retrieval.

2.7.4.2 Software requirements

1. Backend: Python with TensorFlow/PyTorch and OpenCV.
2. Frontend: A cross-platform framework such as Flutter or React Native.

3. Database: A relational (e.g., PostgreSQL) or NoSQL (e.g., MongoDB) database.
4. Cloud Platform: Cloud services like AWS or Google Cloud for hosting and storage.

2.8 Communication Management

The communications management ensures all stakeholders, including the project team and supervisors, have the information they need to perform their roles effectively. Clear and timely communication is essential for the project's success.

Communication Objectives

- Adequate: Provide the right information in the right format.
- Specific: Tailor information for the targeted audience.
- Sufficient: Give all the necessary information without overwhelming details.
- Concise: Keep communication brief and to the point.
- Timely: Address issues and updates at the appropriate time.

Communication Media

- Email: For formal communication, documentation, and key decisions.
- Phone Call/Chats: For quick, informal communication and urgent matters.
- Meetings: For detailed discussions, progress updates, and problem-solving.

2.9 Commercialization aspects

The commercialization strategy for this project is centered on offering a highly specialized and impactful solution to a broad market. The target audience includes individuals focused on personal health and fitness, as well as businesses in the health, wellness, and food

industries. The core value proposition of our system providing real-time, personalized dietary feedback is a unique feature that sets it apart from existing generic fitness and food-logging applications.

We envision a tiered commercialization model:

1. Direct-to-Consumer: The mobile application will be available on app stores (e.g., Apple App Store, Google Play Store) with a freemium model. A basic version with essential food recognition and nutritional analysis will be free, while a premium subscription will unlock advanced features. Premium features would include:
 - a. Personalization Engine: The personalization engine utilizes an Integer Linear Programming (ILP) framework to dynamically select meals that strictly align with a user's unique caloric targets and ingredient constraints.
 - b. Advanced Tracking: The ability to track meals over time and see trends in nutritional intake.
 - c. Expanded Database: Access to a more extensive database of food items, including diverse and regional cuisines.
2. Business-to-Business (B2B): The AI models and backend API can be licensed to other companies on a subscription basis. This would allow businesses, such as food delivery services, healthcare providers, or wellness platforms, to integrate our technology into their own applications. For example, a food delivery service could use our API to instantly provide nutritional information and healthy meal swaps for their menu items.

The project's commercial viability is strong due to its ability to address a critical market need for effective and personalized health solutions. The use of a simple, mobile-based platform ensures a low barrier to entry for users, making the technology highly scalable and widely accessible.

2.10 Testing and Implementation

The FitNourish.AI system was implemented as a modular AI-powered platform integrating personalized nutrition prediction, dynamic meal generation, visual ingredient identification, smart grocery list optimization, nearest store recommendation, predictive health alerts, and progress tracking. Each module was developed independently and then integrated through standardized APIs and data flows to support an end-to-end process from user health profiling to nutrition planning, grocery purchasing, and health monitoring.

The implementation followed a data-driven pipeline consisting of data collection, preprocessing, model development, prediction, optimization, and output generation. User demographic, physiological, health-related, and activity data were processed for nutrition prediction and health risk analysis, while supermarket product and price data from sources such as Keells and Cargills were collected, cleaned, normalized, and stored for grocery optimization.

The nutrition module predicts personalized calorie and macronutrient requirements, while the meal generation and visual ingredient identification components support meal planning and ingredient-based recommendations. The grocery optimization module performs product matching, price comparison, availability checking, and travel-aware store recommendation using product cost, store location, distance, and travel cost. The health tracking module predicts risk levels, provides dietary recommendations, sends reminders, and displays progress through dashboard-based visualizations.

Testing was conducted at both module and system levels to verify functionality, accuracy, reliability, usability, and integration. Functional testing covered user profile handling, nutrition prediction, meal generation, ingredient identification, product matching, price comparison, store recommendation, health risk prediction, dietary recommendation, notifications, and progress tracking.

Model evaluation was performed using suitable metrics. Regression metrics such as MAE, RMSE, and R^2 were used for nutrition prediction, while classification metrics such as accuracy, precision, recall, and F1-score were used for health risk prediction, dietary recommendation, and product matching. Integration testing verified data exchange between the frontend, backend, machine learning models, database, and APIs.

System testing was carried out using realistic user scenarios to confirm that users could enter health details, receive nutrition targets, generate meal and grocery recommendations, compare prices, identify suitable stores, receive alerts, and monitor progress. Overall, the implementation and testing confirmed that FitNourish.AI functions as an integrated, accurate, and practical AI-powered health and grocery optimization system suitable for the Sri Lankan retail and health management context.

3 RESULTS AND DISCUSSION

3.1 Results

The integrated AI-powered systems were evaluated across multiple components using the methodologies outlined in Chapter 2. Performance metrics were computed on held-out test sets, validation splits, and real-world simulation scenarios where applicable. Key results are summarized below.

Product Matching and Grocery Optimization (Smart Grocery System) The hybrid product matching model (Random Forest + rules + canonical mapping) achieved strong performance on the labeled Keells-Cargills dataset (~225 samples after canonical grouping and feature engineering):

- **Accuracy:** 92.3%
- **Precision:** 94.1% (prioritizing avoidance of false matches)
- **Recall:** 89.7%
- **F1-Score:** 91.9%

Random Forest outperformed Logistic Regression (baseline ~81% accuracy) and MLP (~90% accuracy) due to better handling of noisy retail text.

For optimization:

- Single-store recommendations reduced total cost (items + estimated travel) by **18–27%** on average compared to random store selection across 50 simulated user lists.
- Multi-store TPP heuristic (with net-savings threshold) yielded additional **4–9% savings** when beneficial, with average computation time < 2 seconds.
- Haversine + Google Distance Matrix hybrid travel modeling aligned within 8% of real-route costs in validation tests.

Health Risk Prediction and Dietary Recommendation (HPT-PHAS)

- **XGBoost Risk Classifier** (3-class: Low/Medium/High): Test accuracy **96.07% ± 0.32%**, weighted F1-score **96.1%**, AUC **0.9929**. Confusion matrix showed excellent separation, with minimal high-risk false negatives.
- **Voting Ensemble Dietary Recommender** (4-class: Low-Carb, Low-Sodium, Balanced, Mediterranean): Test accuracy **66.94%** post-SMOTE balancing. This outperformed single models (best RF ~65.7%) while maintaining low inference latency (< 250 ms).
- Longitudinal tracking dashboard successfully rendered trends for 365+ days with sub-500 ms query times. Notification delivery success rate exceeded 98% in simulated Android tests.

Personalized Nutrition Prediction (Multi-Disease Adaptive Nutrition) MultiOutput XGBoost (wrapped for 4 targets: calories, protein, carbs, fat) delivered:

- **MAE** (across outputs): 8–12% relative error on test set.
- **RMSE**: Highest for calories (~180–220 kcal deviation).
- **R²**: 0.87–0.94 per target (strongest for calories). SHAP explanations consistently highlighted BMI, activity ratio, age, and heart-rate features as top contributors, aligning with domain knowledge. Cross-validation confirmed generalization (minimal overfitting).

Dynamic Meal Generator & Visual Ingredient Identifier (FitNourish AI)

- **EfficientNetV2B0 Multi-Label CNN** (75 ingredients): Top-1 accuracy **~79–85%** (test set, post-augmentation), with strong multi-label F1 for dominant ingredients.
- **Portion-Independent Regression** (macronutrients): MAPE for calories **~14–16%**, competitive with Nutrition5k benchmarks. 4-4-9 rule validation reduced inconsistencies by ~22%.
- **ILP Personalization Engine**: **84.58%** planning success rate (meeting caloric targets + hard constraints) within < 5 seconds.

Overall System Integration End-to-end latency (image/grocery list → recommendation) averaged 4–12 seconds on standard hardware/cloud. Snapshot-based architecture ensured reproducibility, with historical price/risk trends tracked accurately over multiple scraping cycles.

3.2 Research Findings

The results validate the core hypotheses:

1. Hybrid approaches (ML + rules + canonical) significantly outperform single techniques in handling heterogeneous retail/health data.
2. Travel-aware TPP heuristics and $\text{cost} = \text{items} + \text{travel}$ modeling produce practically superior recommendations compared to price-only baselines.
3. XGBoost-based models excel in healthcare prediction tasks, delivering >95% accuracy for risk classification while supporting explainability via SHAP.
4. Vision + optimization pipelines (EfficientNetV2B0 + ILP) enable dynamic, image-driven meal planning with actionable personalization.
5. Snapshot-based data architecture and layered preprocessing are critical for reproducibility and temporal analysis in dynamic retail/health domains.

Feature engineering (token overlap, unit standardization, risk scores, portion-independent ratios) and ensemble/hybrid strategies were key drivers of performance gains. Ethical safeguards (consent, disclaimers, data minimization) were successfully implemented without compromising utility.

3.3 Discussion

The achieved results demonstrate the viability of integrating data engineering, machine learning, computer vision, and optimization into cohesive AI systems for everyday health and grocery decision support. The grocery optimizer effectively addresses real-world

challenges like product inconsistency and travel costs in the Sri Lankan context (Keells/Cargills), potentially saving users 20–35% overall while respecting convenience.

Health and nutrition components align with literature on XGBoost superiority in tabular clinical data and the value of ensemble recommenders, even in challenging multi-class dietary tasks. The 96%+ risk prediction accuracy is particularly promising for proactive alerts, though the moderate dietary accuracy (~67%) reflects the inherent complexity and overlap in lifestyle-based recommendations—suggesting room for richer features (e.g., longitudinal wearables) or user feedback loops.

Visual meal identification on Nutrition5k-level data performs competitively, validating transfer learning and portion-independent techniques. ILP optimization adds mathematical rigor, ensuring recommendations are not just predictive but feasible under constraints. Limitations include dataset scale (e.g., ~225 matching pairs for grocery; Nutrition5k’s controlled setting), potential domain shift in real-home images, and dependency on scraping stability.

Implications: These systems bridge theory and practice, offering scalable tools for precision nutrition, cost-efficient shopping, and preventive health in resource-constrained settings like Sri Lanka. Future work should focus on larger longitudinal datasets, federated learning for privacy, real-user RCTs for behavioral impact, and expanded retailer integration. The modular, snapshot-driven design facilitates ongoing improvement and commercialization (freemium app + B2B API licensing).

Overall, the research confirms that thoughtfully combined AI pillars—entity resolution, XGBoost ensembles, CNNs, ILP, and explainable techniques—can deliver tangible value while maintaining transparency, efficiency, and ethical standards.

4 CONCLUSION

This research presented FitNourish.AI, an AI-powered Smart Grocery and Personalized Health Optimization System designed to support users in making healthier, personalized, and cost-effective food-related decisions. The system integrates four major components: multi-disease adaptive nutrition prediction, dynamic meal generation with visual ingredient identification, smart grocery list optimization with nearest store recommendation, and predictive health alert with progress tracking.

The proposed system addresses the limitations of fragmented nutrition, health monitoring, and grocery recommendation applications by combining them into a unified platform. Through machine learning models, the system predicts personalized calorie and macronutrient requirements, identifies health risk levels, recommends suitable dietary patterns, and supports continuous health progress monitoring.

In addition, the grocery optimization component connects personalized nutrition needs with real-world supermarket data. By matching products across retailers, comparing prices and availability, and considering travel distance and travel cost, the system provides practical grocery purchasing recommendations. This makes the solution more useful in real-life situations, especially within the Sri Lankan retail context.

Overall, the project achieved its main objective by developing an integrated AI-based system that supports personalized nutrition planning, health-aware decision-making, and economical grocery purchasing. Future improvements can include expanding supermarket data sources, improving real-time data updates, enhancing visual food recognition accuracy, integrating more wearable devices, and validating the system with a larger group of real users.

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